The landscape of recursive self-improvement

Foundations, enabling mechanisms and emerging systems

Representative studies

Improvement mechanisms

Update targets

Read from the center outward

Affiliation marks

One selected institution per study

RSI

Update loop

Weights

Tasks

Memory

Programs

Feedback

Improver

Propose tasks and co-evolve

Generate, filter and train

Accumulate and reuse experience

Search, edit and execute

Judge outputs and refine judges

Revise the update or research process

SEAL 2025

ReST 2023

Self-Instruct 2022

STaR 2022

AgentEvolver 2025

R-Zero 2025

Absolute Zero 2025

AgentGym 2024

POET 2019

ACE 2025

A-MEM 2025

ExpeL 2023

Voyager 2023

Reflexion 2023

GEPA 2025

DGM 2025

ADAS 2024

DSPy 2023

OPRO 2023

CREAM 2024

Self-Taught Evaluators 2024

Meta-Rewarding 2024

Self-Rewarding 2024

CRITIC 2023

Promptbreeder 2023

Metaⁿ 2026

Ind. Bilevel Autoresearch 2026

Hyperagents 2026

STOP 2023

AI Scientist 2024